Maurizio Ungaro

Biography

Staff Scientist, Jefferson Lab | Nuclear physicist and simulation software developer
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Short Bio

Maurizio Ungaro is a nuclear physicist and simulation software developer in Jefferson Lab Hall-B. His work spans GEMC and GEMC3, Geant4-based detector simulation, CLAS12 systems, and nucleon-structure research.

Medium Bio

Maurizio Ungaro is a Staff Scientist in Jefferson Lab Hall-B, working at the intersection of nuclear physics, detector systems, and scientific software. He develops GEMC and GEMC3, Python tools through pygemc, and CLAS12 simulation workflows. His recent work includes CAD geometry repair and validation, reproducible performance studies, and interactive visualization. He supports distributed simulation production on the Open Science Grid and works on the CLAS12 Low Threshold Cherenkov Counter. He also writes tutorials and technical documentation for researchers and students.

Long Bio

Maurizio Ungaro is a nuclear physicist and simulation software developer at Jefferson Lab. His research focuses on the internal structure and dynamics of the nucleon, including meson electroproduction, nucleon resonances, and the transition between hadronic and partonic degrees of freedom. His technical work connects detector operation, Geant4 simulation, CLAS12 production workflows, and scientific software infrastructure.

He develops and supports GEMC, a database-driven Geant4 simulation framework, and its next-generation GEMC3 framework with Python tools through pygemc. Recent development includes CLAS12 geometry and detector-plugin migration, CAD mesh repair and optimization, and reproducible construction, overlap, and particle-tracking checks. He also works on interactive detector visualization, including Geant4 XR integration, multilingual software tools, and portable build and installation workflows.

His production work includes CLAS12 simulation releases and simGrid workflows using HTCondor and Open Science Grid resources. His detector responsibilities include operation, maintenance, calibration, and performance studies of the CLAS12 Low Threshold Cherenkov Counter. Through Geant4 at JLab, tutorials, examples, and technical notes, he helps researchers and students use and validate detector simulations.

His earlier and collaborative work includes experiments at SPring-8 and participation in the Heavy Photon Search collaboration.

ORCID / LinkedIn Draft

ORCID biography: Staff Scientist at Jefferson Lab working on nuclear physics, Geant4 simulations, GEMC and GEMC3, pygemc, CLAS12 detector systems, validated CAD geometry, and distributed scientific computing workflows.

LinkedIn headline: Nuclear Physicist | Geant4 and GEMC Simulation Developer | CLAS12 Software and Detector Systems | Jefferson Lab